

Lecture 20

3/6/26
JRA

2P $\hat{=}$ Deep Imaging (cont)

Two P involves exciting w/ 2 photons

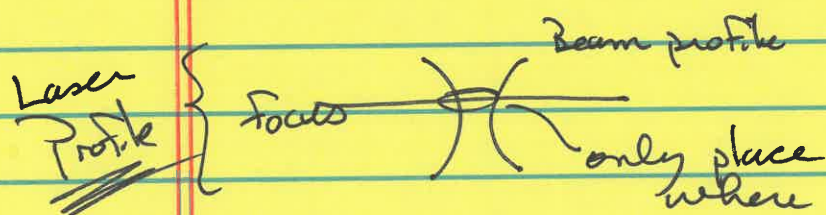
(1) Excitation light has 2×2 of 1 p photon

1p 500nm $\rightarrow$ 2p 1000nm

(2) Intense

2P excitation will only take place if two photons are absorbed simultaneously

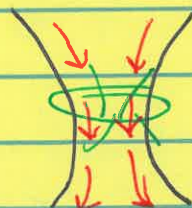
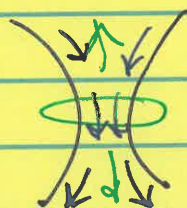
Intense IR light from a ^{focused} laser confines excitation & emission laser focus



Total excitation

(3) No pinhole is required ^{because there is} - No OOF Fluorescence

Compare w/ LSCM because both are scanning techniques —



Excite only here

(4) Need to scan like LSCM

(5) Advantages - deep images

Thin sample $\rightarrow$ use LSCM

Thick sample $\rightarrow$ use 2P

(a) Reduced attenuation of excitation light

DEMO

Green ^(λ blue) light - doesn't penetrate thumb
because absorbed by
endogenous
molecules

Red ^(λ IR) light - penetrates
& doesn't
get deep
enough to
excite

(b) Reduced scattering problems

Looked at tissue slide

Confocal

2P

Also tissue
samples



Bad OOF photons that should be rejected
by the pinhole can scatter into the pinhole

⇒ extra blur

In focus photons that should go thru pinhole
can scatter & miss

⇒ lost good signal

Overwhelming problems > ~50 μ m deep

⇒ LSCM users

2P no pinhole ⇒ massively reduced
scattering problems

LSCM - bleach everywhere
photo B photo T everywhere

2P - NOT true only at focus

(c) 2P reduced photo B & photo T because only
focal plane excited unlike LSCM

Looked at ex & em spectra

Aside

1P ex shorter λ
2P ex longer λ

Disadvantages

- (1) Need super intense (pulsed) ^{IR} lasers (expensive)
- (2) H_2O absorbs in IR
- (3) Need to scan
- (4) Longer $\lambda \Rightarrow$ ^{worse} poorer resolution $\sqrt{2}$

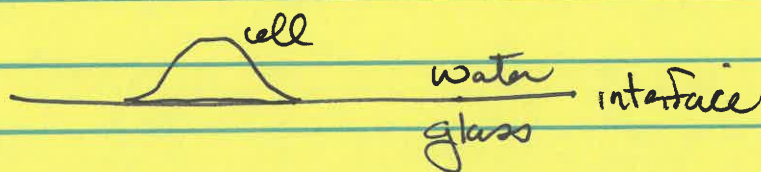
TIRFM

How about a ^{technique} surface imaging? TIRFM

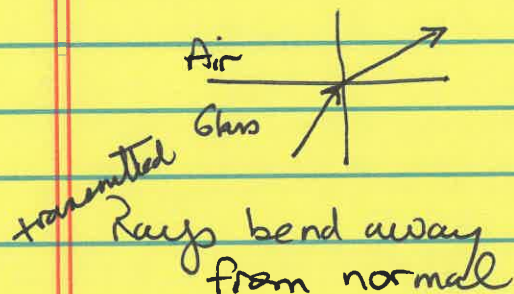
Why? exocytosis
endocytosis
protein mobility in membranes

Intro to the essential idea - ray bending at an interface

Idea behind
fiber optic
cables

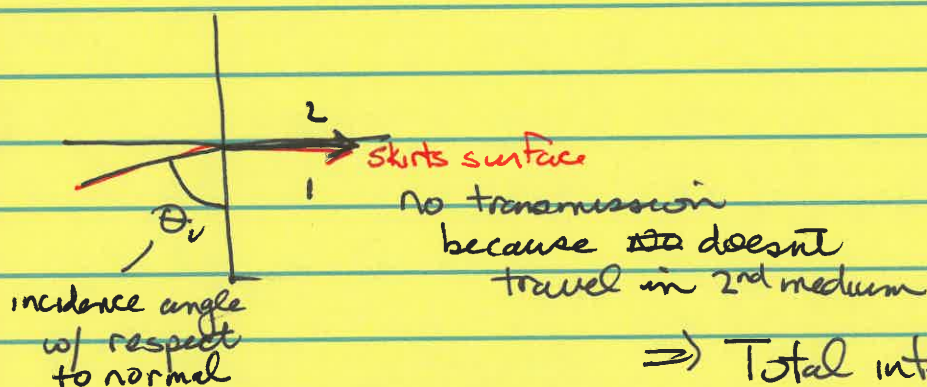


Looked back at discussion of oil immersion objectives



Consequences of Snell's Law

Use oil immersion because otherwise you lose signal ϵ , so lose resolution



$\Rightarrow$ Total internal reflection

transmitted/refracted ray bends to 90°

What is the "critical angle" (critical incidence angle) that creates TIR?
Angle at which TIR occurs

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

Snell's Law

1 = glass
2 = air

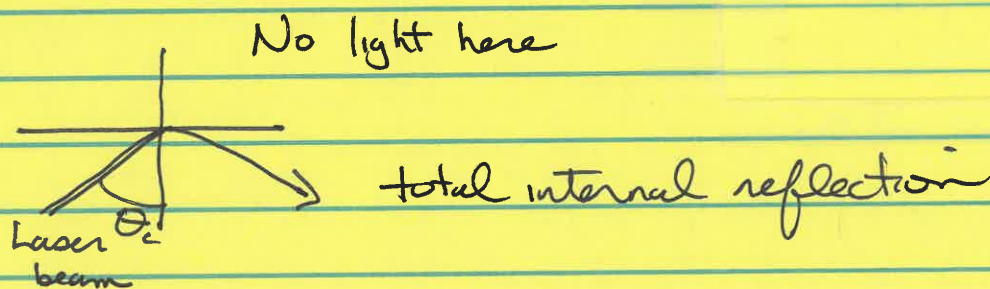
$$n_{\text{glass}} \sin \theta_c = n_{\text{air}} \sin \theta_2$$

$$1.5 \sin \theta_c = 1 \sin 90^\circ = 1$$

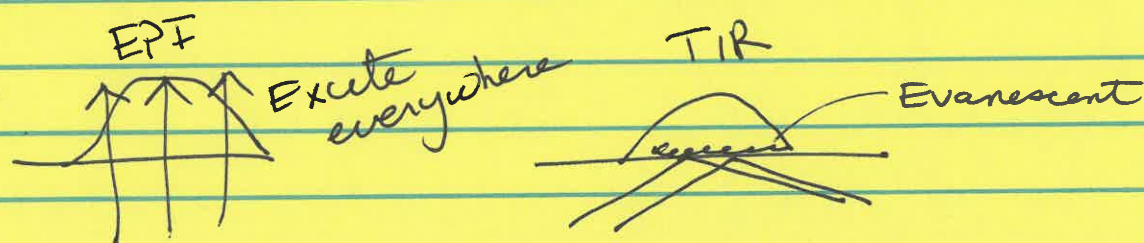
$$\sin \theta_c = \frac{1}{1.5}$$

$$\theta_c = \arcsin\left(\frac{2}{3}\right)$$

$$= 41.8^\circ \approx 42^\circ$$



$$\theta_c \geq 62.5^\circ$$



Geometric optics — no penetration
Wave optics — evanescent field

Wave optics modifies the predictions of ^{geometric} ray optics (Snell's Law) to say an evanescent field in second medium
 * short lived in distance (cell with fluorescent)

Buy Idea: ψ field exists $\oplus$ only near surface

Looked at PP slides: penetrates about 100 nm

only exists at surface

Markedly reduced OOF fluor

Not PhotoB / PhotoT

ANIMATION

Mol Exp



MOVIE

Granules at cell surface
 showed exocytosis